

AKSHIT NANDA

PROFILE

Motivated, result-driven, and passionate towards computer science with experience in various domains of this exciting field. Highly qualified, possessing a bachelor's and master's degree in mathematics with minor in data science.

EDUCATION

DAV PUBLIC SCHOOL, Kalinga Nagar, Bhubaneswar, Odisha

Upto class - 12

Class 10 : CGPA 10

Class 12 : 89 % (Mathematics-98 , Chemistry-96 , Physics 84)

Indian Institute of Science Education and Research Mohali, Punjab

2018 - 2023

CPI = 6.6 (Bachelors and Masters in Mathematics with a minor in Data Science)

SKILLS

Programming Language: Python, SQL

Software / Tools: Latex, CDO, MS Word, MS Excel, MS Powerpoint

PROJECTS

1. Summer Project **2018**

Dr. P. Balanarayan

Solving Schrodinger Equation for Hydrogen atom in Morse potential

My interest in programming made me select this project which involved solving Solving Schrodinger Equation for Hydrogen atom in Morse potential using python.

2. Semester Project **2021**

Dr. Sarabjot Singh Anand

News Recommender System

(a) Web scrapped to collect data and create a large database.

(b) Used NLP to make an algorithm that takes user experience and recommend related news articles to the user.

3. Lab Project **2022**

Dr. Raju Attada

Cold Stress in Northern India

(a) Used 22 years of Climate data to for data analysis.

(b) Studied the pattern of cold stress in the Indian subcontinent and used the data to predict cold stress for future years.

4. **Master's Thesis** **2022 - 2023**
Dr. Sarabjot Singh Anand
Improve Algorithmic Fairness in Recidivism Prediction
(a) Used the Recidivism dataset to calculate accuracy and fairness.
(b) Studied different bias mitigation algorithm to optimize the trade-off between accuracy and fairness.
5. **Project** **Ongoing**
Dr. Raju Attada
Create a Pipeline for Extreme Climate Indices
(a) Contributed to the development of precipitation indices and data analysis
(b) Created a pipeline to optimize the memory allocation to support big data.
(c) Details can be found [here](#).

COURSES

IISER Mohali Courses

1. **Mathematics:** Probability and Statistics; Analysis in R_n ; Integers, polynomials and matrices; Ordinary and partial differential equation; Discrete mathematics; Measure and probability; Advanced probability; Number theory; Random graphs
2. **Data science:** Introduction to data science; Machine learning; Nonlinear dynamics, chaos and complex systems; Climate data analysis and visualization
3. **Biology:** Cellular Basis of Life; Gene Expression and Development; Genetics and Evolution; Behaviour and Ecology
4. **Physics:** Mechanics; Electromagnetism; Waves and optics; Thermodynamics and Statistical Physics
5. **Chemistry:** Chemistry of Elements and Chemical Transformation; Atom Molecules and Symmetry; Spectroscopy and other Physical Methods; Energetics and Dynamics of Chemical Reactions

Online Certification

1. Python for Data Science
2. Basic Statistics
3. Mathematics for Machine Learning: Linear Algebra
4. Learning SQL Programming
5. AI Engineering Professional

ACADEMIC ACHIEVEMENTS

1. Bagged rank 28 in JUNIOR MATHEMATICS OLYMPIAD.
2. Selected for a week discussion meeting on Data science probabilistic optimization methods held at International Center for Theoretical Sciences TIFR.
3. Selected for a two-week internship program in INFOSYS under the "Catch them young" program.
4. Selected as a Gauth expert tutor. (Online platform to clear academic doubts of students).
5. Several Olympiad medals in Science and Math department.